# Lesson 13: A Hit That Reacts

Scratch

VIDEO GAME CREATION

**This week: getting hit finally LOOKS like getting hit. A brief flash, then a small explosion, before the enemy disappears.**

**Open your own [YourName]Game project from Lesson 12 - don't start a new one.**

*Important: your score, lives, wave, and win/lose logic should all still work exactly as before. Today only changes how the enemy's removal LOOKS, not when your game-state actually changes.*

## Setup

1. Open your [YourName]Game project from Lesson 12.
2. Add the explosion_placeholder.png costume to a brand-new sprite named Explosion (Scratch will let you draw/import a new sprite from a file).
3. Import enemy_invader_white.png (this lesson's Assets folder) as a NEW COSTUME on the Enemy sprite itself - not a new sprite. In the Costumes tab, use the upload/import costume option so it becomes a second costume alongside the Enemy's existing red one. Rename the two costumes clearly - e.g. normal and flash - so the switch-costume blocks below are unambiguous.
4. Create two new variables, for all sprites: hit_x and hit_y. Scratch's broadcast doesn't carry any data along with it, and "go to [Enemy]" would go to the Enemy sprite's own template position, not the specific clone that actually got hit - these two variables are how the Explosion sprite finds out WHERE to appear.
5. Add "switch costume to [normal]" to the Enemy's existing "when I receive [spawn enemies]" spawn script (or wherever each enemy resets), so a fresh enemy never starts already showing its flash costume.

## Level 1 - Minimal (MVP)

*On every hit, the enemy briefly flashes, and a small independent explosion effect appears and cleans itself up. Score, lives, wave, and win/lose logic are all unaffected.*




*Everything in the top box happens the instant the hit is detected. Only the enemy's own removal - the very last step - waits for the pause to finish.*

6. On the Explosion sprite, add a separate script: "when green flag clicked" -> hide. This guarantees Explosion starts hidden every time you play, regardless of whatever visibility state got left over from editing - more reliable than just toggling the eye icon in the Sprite pane, since that's easy to forget or to accidentally flip back on later.
7. On the same sprite, add a second new script: "when I receive [explosion]" -> show -> go to x: (hit_x) y: (hit_y) -> wait 0.2 seconds -> hide. hit_x/hit_y (set by the Enemy clone right before broadcasting) are what make this work correctly even with multiple Enemy clones on screen - "go to [Enemy v]" would always go to the Enemy template sprite's own position, never whichever clone actually got hit.
8. In the Enemy's touching-Bullet branch, right after the existing "start sound [sfx_hit]" line, add: "set [hit_x] to (x position)" and "set [hit_y] to (y position)" (captures THIS clone's own position, using Scratch's built-in x position/y position reporters), then "broadcast [explosion]" (spawns the visual independently, with zero effect on anything else) and "switch costume to [flash]" - all four run immediately, same as the sound.
9. Switching to the flash costume is a genuine costume swap, not a color effect - "set [color] effect" in Scratch shifts HUE around the color wheel rather than blending toward white, so it never actually produces a white flash on a colored sprite. Swapping to the enemy_invader_white.png costume imported in Setup is the only way to get a genuinely white flash here.
10. Your existing "broadcast [enemy hit]" stays exactly where it already is, right after those new lines - GameManager still hears about the hit, and still updates

score/enemies_remaining/wave-check, at exactly the same instant as last week. Nothing about that broadcast changes.

11. Add "wait 0.1 seconds" AFTER broadcast [enemy hit], then move "delete this clone" to run after that wait - this is the only thing actually delayed: the enemy sticks around, visibly flashed, for a moment before it's actually removed. No need to switch back to the normal costume first - the clone is deleted a moment later anyway.
12. Click the stage and play through a full round - each hit should flash the enemy white briefly, and a small burst should appear at that spot for a moment.

*Highlighted lines are new this week; everything else is code you already have.*

```
Enemy's touching-Bullet branch (revised):
if <touching Bullet?> then
  start sound [sfx_hit]
  set [hit_x] to (x position)
  set [hit_y] to (y position)
  broadcast [explosion]
  switch costume to [flash]
  broadcast [enemy hit]
  wait 0.1 seconds
  delete this clone

Explosion sprite's own scripts (NEW - two separate scripts):
when green flag clicked
hide

when I receive [explosion]
show
go to x: (hit_x) y: (hit_y)
wait 0.2 seconds
hide
```

## Level 2 - Complete (Alpha/Beta)

*The flash and explosion play correctly on every single hit, with no leftover explosions, no enemies stuck mid-flash, and no change to score/lives/wave timing.*

- Play a full round: every single hit shows a flash and an explosion, with no enemy getting visually "stuck" flashed.
- Confirm score, wave-clear, and win/lose all still happen at exactly the moments they did last week - broadcast [enemy hit] hasn't moved at all, only delete this clone is delayed.
- Confirm the broadcast-based Explosion sprite never blocks or delays anything else - broadcasts don't wait for their receivers to finish.
- Shoot two enemies in different spots in quick succession and confirm each explosion appears at the RIGHT enemy's position, not both appearing at the same spot - this is exactly what hit_x/hit_y are for.

## Level 3 - Refined (Ready to Publish) - pick any that interest you

- Visual: instead of one flat 0.1-second flash, alternate between the flash and normal costumes a couple of times (e.g. flash, normal, flash) with short waits between - try around 0.1 seconds each, fast enough to read as a flicker but slow enough to actually be visible. Since Enemy only has these two costumes (normal and flash, in that order from Setup), you can swap the single "switch costume to [flash]" for three "next costume" blocks with a wait after each of the first two - each clone always starts on normal, so

next costume just toggles it back and forth: flash, (wait), normal, (wait), flash, then straight into broadcast [enemy hit]. This only works as a clean toggle because there are exactly two costumes in that order - if Enemy ever gains a third costume later, go back to explicit "switch costume to [name]" blocks instead.

- Visual: grow the Explosion as it plays - "set size to (100) %" right when it appears (so leftover size from a previous explosion doesn't carry over, since this is the same Explosion sprite reused every time, not a fresh clone), then step up to 105%, 110%, 115% with the same short waits you'd use for a flicker, for a simple expanding-burst look - the same idea as the explosion-scaling stretch goal in every other engine this week. (If you'd rather hand-animate instead: give the Explosion sprite 2-3 different costumes and cycle through them with "next costume" during its 0.2-second lifetime - duplicate the costume in the Costumes tab and edit each copy in the Paint Editor - but the size-based approach needs no extra art at all.)
- Structure: notice the Explosion sprite knows nothing about score, lives, or waves - it does exactly one job, triggered independently of the rest of the hit-handling logic.

## If Something's Wrong

- Enemy vanishes instantly, no flash visible: confirm "wait 0.1 seconds" was actually added, and that "delete this clone" was moved to AFTER it, not before.
- Enemy flashes some other color instead of white, or doesn't visibly change at all: confirm the flash uses "switch costume to [flash]", not "set [color] effect to 100" - the color effect shifts hue rather than blending toward white, so it was never capable of producing a true white flash here.
- Explosion never appears, or always appears in the same wrong spot: confirm hit_x/hit_y are being set (from the Enemy clone's own x position/y position) immediately BEFORE broadcast [explosion], and that Explosion's own script uses "go to x: (hit_x) y: (hit_y)", not "go to [Enemy v]" - the latter always goes to the Enemy template sprite's own position, never the specific clone that was actually hit.
- Score changes twice per hit, or win-check fires early: confirm "broadcast [enemy hit]" wasn't accidentally duplicated or moved when adding the new lines - it should still appear exactly once, in its original spot.

## Want More?

Student Resource List > Scratch section, or scratch.mit.edu/ideas